AC Type-2 Charging - II

Introduction

Imagine plugging in your electric vehicle (EV) for a charge. While the process seems simple, a sophisticated communication dance occurs behind the scenes to ensure safe and efficient power transfer. This lecture delves into the intricate details of AC Type-2 charging, building upon our previous discussion of the Proximity Pilot (PP) circuit. We will explore how the Control Pilot (CP) circuit facilitates communication between the Electric Vehicle Supply Equipment (EVSE) and the EV, orchestrating the charging sequence through a series of defined states and signals. Understanding these protocols is crucial for comprehending the intelligence embedded within modern EV charging infrastructure, where safety, compatibility, and controlled power delivery are paramount.

Core Concepts

Term	Definition	Significance
`PP circuit`	A circuit within the EV connector that uses a variable resistor (R6) to signal the EV about the connected plug's current capacity and type (AC or DC).	Ensures the EV identifies the correct charging plug and its maximum current rating, preventing overcurrent or incompatible connections.
`Control Pilot (CP)`	A dedicated communication line between the EVSE and EV that uses a Pulse Width Modulation (PWM) signal to exchange vital information.	Facilitates low-level communication, indicating readiness for charging, maximum current available, and other status updates.
`PWM`	Pulse Width Modulation, a signaling technique where the width of a pulse is varied to convey information. In AC charging, it's a 1 kHz signal.	The primary communication method on the CP line, conveying EVSE readiness and maximum current capacity through its duty ratio.
`Duty Ratio`	The fraction of time within a PWM cycle during which the	Communicates the maximum current the EVSE can safely deliver to the EV, allowing the

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	positive voltage level is applied (t_on / T).	EV's Battery Management System (BMS) to adjust its demand.
`EVSE`	Electric Vehicle Supply Equipment, commonly known as a charging station or charger.	Responsible for generating the PWM signal, monitoring the CP line voltage, and providing AC power to the vehicle.
`EV`	Electric Vehicle.	Interprets the PWM signal's duty ratio and frequency, changes the CP line voltage level to signal its readiness, and manages onboard power conversion.
`Charging States`	A series of predefined operational modes (A, B1, B2, C1, C2, D, E, F) that the EVSE and EV transition through during a charging session.	Defines the sequence of events from plug connection to charging completion or error, ensuring a standardized and safe charging process.

Detailed Analysis

The AC Type-2 charging process relies on a sophisticated communication protocol, primarily governed by the Control Pilot (CP) circuit, to ensure safe and efficient energy transfer. This protocol is standardized by SAE J1772 and IEC 61850.

1. Proximity Pilot (PP) Circuit Recap

Before CP communication begins, the Proximity Pilot (PP) circuit plays a crucial role in identifying the connected charging plug.
- The PP circuit contains a resistor, R6, whose value changes based on the maximum current capacity and type (AC or DC) of the connected plug.
- For instance, a 1500-ohm R6 indicates a 13-ampere AC plug, resulting in a voltage of 3.87 volts at the PP pin relative to the protective earth (PE).
- The EV's control system senses this voltage, allowing it to determine the plug's capabilities (e.g., 63A, 32A, 20A, 13A) and confirm it's a correct and compatible connector.
- Once the correct plug is identified, the vehicle immobilizes, and the communication shifts to the Control Pilot (CP) line.

2. Control Pilot (CP) Circuit Communication

The CP circuit facilitates low-level communication using a 1 kHz Pulse Width Modulation (PWM) signal.
- The EVSE generates this 1 kHz PWM signal, which has a peak positive voltage of +12V and a negative voltage of -12V.
- The EV measures the duty ratio and frequency of this PWM signal. If the frequency is not within the required range, communication stops.
- The EV, in turn, changes the voltage level (amplitude) of the PWM pulses by controlling its internal S2 switch. This change in voltage level is measured by the EVSE, informing it of the EV's readiness.
- Unlike DC charging, where complex high-level communication is needed due to power conversion happening in the EVSE, AC charging's power conversion occurs within the vehicle's onboard charger. This simplifies the communication to a PWM-based system, as the EV's Battery Management System (BMS) primarily controls the charging process internally.

3. CP Circuit Parameters

The CP circuit operates within specific electrical parameters to ensure reliable communication:
- **Positive Idle Voltage (Uoch)**: +12V (+/-0.6V)
- **Negative Idle Voltage (Uocl)**: -12V (+/-0.6V)
- **Frequency (f)**: 1000 Hz (+/-0.5 Hz)
- **Maximum Rise Time (Trg)**: 2 microseconds (10% to 90%)
- **Maximum Fall Time (Tfg)**: 2 microseconds (10% to 90%)
- **Minimum Settling Time (Tsg)**: 3 microseconds
- **Resistance of Source (R1)**: 1000 ohms (+/-3%)
- **Recommended Capacitance for Interference Suppression (Cs)**: 300 pF
- **Maximum Total Line Capacity (Cs + Cc)**: 3100 pF

4. Charging Sequence States

The charging process progresses through several defined states, each indicated by specific voltage levels on the CP line, measured by the EVSE, and the activation status of the PWM signal. The circuit involves an S1 switch (controlled by EVSE) and an S2 switch (controlled by EV), along with resistors R1 (1kΩ), R3 (2.74kΩ), and R2 (1.3kΩ or 270Ω). A diode D1 ensures only positive voltage passes through for measurement.

- **State A: Not Connected**
- **VDC**: +12V

- **EVSE Status**: S1 connected to fixed +12V, oscillator off (PWM deactivated).
- **EV Status**: Not connected to EVSE.
- **Significance**: The EVSE detects +12V, indicating no vehicle is plugged in.
- **State B1: EV Connected, Not Ready**
- **VDC**: +9V (calculated via voltage divider: (R3 / (R1 + R3)) * 12V = (2740 / (1000 + 2740)) * 12V ≈ 8.79V ≈ 9V)
- **EVSE Status**: S1 still connected to fixed +12V, oscillator off (PWM deactivated).
- **EV Status**: Plugged in, but S2 switch is open (not connected to R2).
- **Significance**: EVSE detects +9V, recognizing a vehicle is connected but neither EVSE nor EV is ready for energy transfer. EV detects PWM is off, meaning EVSE is not ready.
- **State B2: EV Connected, EVSE Ready**
- **VDC**: +9V (positive peak), -12V (negative peak)
- **EVSE Status**: S1 connected to the 1kHz oscillator (PWM activated).
- **EV Status**: Plugged in, S2 switch is open.
- **Significance**: EVSE generates PWM. EV detects the PWM frequency and duty cycle, understanding that the EVSE is now ready for energy transfer. The voltage level remains +9V (positive peak) because D1 only passes positive voltage, and S2 is still open.
- **State C2: EV and EVSE Ready (Charging)**
- **VDC**: +6V (positive peak), -12V (negative peak)
- **EVSE Status**: S1 connected to the 1kHz oscillator (PWM activated).
- **EV Status**: S2 switch is closed, connecting R2 (1.3kΩ) in parallel with R3 (2.74kΩ).
- **Significance**: The parallel combination of R2 and R3 (approx. 882Ω) changes the voltage divider. VDC = (R_parallel / (R1 + R_parallel)) * 12V = (882 / (1000 + 882)) * 12V ≈ 5.79V ≈ 6V. EVSE detects +6V, indicating the EV is also ready for charging. This is the state where actual AC power transfer begins.
- **State C1: EV Ready, EVSE Not Ready (Error)**
- **VDC**: +6V
- **EVSE Status**: S1 connected to fixed +12V, oscillator off (PWM deactivated).
- **EV Status**: S2 switch is closed, connecting R2 (1.3kΩ) in parallel with R3 (2.74kΩ).

- **Significance**: EV is ready (indicated by +6V), but EVSE has an error or is not ready (PWM deactivated). EV detects no PWM, stopping charging.
- **State D: EV Ready, Requires Ventilation**
- **VDC**: +3V (positive peak), -12V (negative peak)
- **EVSE Status**: S1 connected to the 1kHz oscillator (PWM activated).
- **EV Status**: S2 switch connects a different resistor (270Ω) in parallel with R3 (2.74kΩ).
- **Significance**: The parallel combination of 270Ω and R3 (approx. 246Ω) changes the voltage divider. VDC = (R_parallel / (R1 + R_parallel)) * 12V = (246 / (1000 + 246)) * 12V ≈ 2.37V ≈ 3V. EVSE detects +3V, indicating the EV is ready but requires ventilation (for older EVs with vented batteries).
- **State E: Fault Condition (Short Circuit)**
- **VDC**: 0V
- **Significance**: A short circuit between the CP and PE pins, or an EVSE error, causes the voltage to drop to 0V. Both EV and EVSE detect this fault and stop energy transfer.
- **State F: EVSE Unavailable/Error**
- **VDC**: -12V
- **Significance**: This state indicates the EVSE is unavailable, unplugged, or has a severe error. The measuring circuit might also have an error.

5. Duty Ratio and Current Control

While voltage levels indicate readiness, the duty ratio of the PWM signal communicates the maximum current the EVSE can provide.
- The EVSE generates the PWM, defining the duty ratio. The EV's BMS interprets this to determine the maximum current it can demand.
- **Duty Ratio Ranges and Max Current:**
- `0% - 3%`: No current allowed.
- `3% - 7%`: Reserved for DC charging (high-level communication).
- `7% - 8%`: No current allowed.
- `8% - 10%`: Max 6 Amperes.
- `10% - 85%`: Max current = 0.6 * Duty Ratio (e.g., 20% duty ratio allows 0.6 * 20 = 12A).
- `85% - 96%`: Max current = (Duty Ratio - 64) * 2.5.
- `96% - 97%`: Max 80 Amperes.
- `97% - 100%`: No current allowed.
- The EV's BMS ensures that the current drawn is always less than or equal to the

maximum current indicated by the EVSE's duty ratio. The EVSE can dynamically change this duty ratio based on its internal conditions (e.g., temperature, grid load, variable pricing).

6. Overall Charging Waveform and Sequence

The entire AC charging process follows a dynamic sequence reflected in the CP pin waveform:
1. **Initial State (A)**: +12V, no plug connected.
2. **Plug-in (B1)**: Voltage drops to +9V (PWM still off). EVSE recognizes connection.
3. **EVSE Ready (B2)**: EVSE activates PWM (positive peak +9V, negative peak -12V). EV detects PWM, knows EVSE is ready.
4. **EV Ready (C2 or D)**: EV closes S2 switch. Voltage drops to +6V (C2) or +3V (D, if ventilation needed), with -12V negative peak. EVSE detects this, knows EV is ready.
5. **AC Power On**: After a delay (TAC on), the contactor closes, and AC voltage appears on L1, L2, L3 pins. The EV's BMS starts drawing current, ensuring it's less than the maximum allowed by the PWM duty ratio.
6. **Current Adjustment**: EVSE can change the PWM duty ratio to adjust the maximum allowed current. The EV responds by adjusting its current demand.
7. **Charging Completion**: When the battery is full, the EV opens its S2 switch, returning the CP voltage to +9V (B2 state). This signals to the EVSE that the EV is no longer ready.
8. **AC Power Off**: After a delay (TAC off), the contactor opens, removing AC voltage from the power pins.
9. **PWM Deactivation**: EVSE deactivates the PWM.
10. **Plug Out**: The charger can now be safely unplugged, returning to State A.

Key Takeaways

- [ ] The Proximity Pilot (PP) circuit identifies the connected plug type and maximum current capacity using a variable resistor (R6) and voltage sensing.
- [ ] The Control Pilot (CP) circuit uses a 1 kHz PWM signal for low-level communication between the EVSE and EV.
- [ ] The EVSE generates the PWM, and the EV measures its frequency and duty ratio to understand EVSE readiness and available current.
- [ ] The EV signals its readiness by changing the CP line's voltage level (e.g., from +9V to +6V or +3V) by controlling its internal S2 switch.

- [ ] Different voltage levels on the CP line correspond to distinct charging states (A, B1, B2, C1, C2, D, E, F), indicating connection status, readiness, or fault conditions.
- [ ] The duty ratio of the PWM signal directly communicates the maximum current the EVSE can safely provide, allowing the EV's BMS to regulate its demand.
- [ ] AC charging involves power conversion inside the vehicle, simplifying communication compared to DC charging, where power conversion occurs in the EVSE.

Conclusion

The AC Type-2 charging process, while seemingly straightforward to the user, is underpinned by a precise and robust communication protocol. The interplay between the PP and CP circuits, the dynamic changes in voltage levels, and the interpretation of PWM duty ratios form a silent dialogue between the EVSE and the EV. This low-level communication ensures that the vehicle is correctly identified, the charging process is initiated safely, and the power delivery is optimized according to both the EVSE's capabilities and the EV's requirements. This foundational understanding of AC charging communication sets the stage for appreciating the greater complexities involved in DC fast charging, where the need for high-level data exchange becomes even more critical due to the external power conversion.